Math 9/10 - Reasoning and Analyzing Units 1+2 (V1)

Name : _____

Teacher's name: _____

Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) The line $y = x$ has a y-intercept of 0.

   (b) If a, b are real numbers, then $\frac{a}{b}$ is a rational number.

   (c) If two lines have slopes m and $-m$ respectively, then they are perpendicular.

   (d) Given two lines in standard form, $Ax + By = C$ and $Dx + Ey = F$, if the two lines are identical, then $A = D, B = E$, and $C = F$.

   (e) The sum of two rational numbers can be an irrational number.

2. A student was asked to write a line equation parallel to $3x + 2y = 6$ and crosses the point $(4, 3)$. Which step did he first make an error? Explain the error.

$$3x + 2y = 6$$

$$2y = -3x + 6 \qquad \textbf{Step 1}$$

$$y = -\frac{3}{2}x + 3 \qquad \textbf{Step 2}$$

$$y - y_1 = m(x - x_1) \qquad \textbf{Step 3}$$

$$y - 3 = \frac{2}{3}(x - 4) \qquad \textbf{Step 4}$$

3. In each of the following, one item doesn't belong....

a) Which number set doesn't belong? Explain.

i. $\mathbb{N}$ ii. $\overline{\mathbb{Q}}$ iii. $\mathbb{Z}$ iv. $\mathbb{W}$

b) Which line doesn't belong?

i. $y = 1$ ii. $y = x$ iii. $3x - 2y = 0$ iv. $x = 0$